Review Questions for Lecture 12

1. Suppose you have a set of 10 objects. How many ways can you order them? Is your answer closer to one hundred, one thousand, one million, one billion, or one trillion?

2. Suppose you have a set of 15 objects. How many ways can you order them? Is your answer closer to one hundred, one thousand, one million, one billion, or one trillion?

3. Suppose you have a set of 10 objects. How many ways can you choose 5 of them? Is your answer closer to one hundred, one thousand, one million, one billion, or one trillion?

4. Suppose you have a set of 25 objects. How many ways can you choose 10 of them? Is your answer closer to one hundred, one thousand, one million, one billion, or one trillion?

5. How many different ways can you write a string of letters that contains 4 "A"s and 7 "B"s? For example, BBAABABBBAB, ABABABABBBB, etc.

6. How many different ways can you write a string of letters that contains 4 "A"s and 7 "B"s, and 3 "C"s? For example, BCBAABCACBBBAB, ABABACCCBABBBB, etc. Hint: This may take some thinking; you can't just apply any of the formulas directly. Another hint: there is more than one way to approach this problem.

7. Suppose a box contains 12 red balls and 8 green balls. Suppose that you pull out a ball at random, check the color, and then replace the ball in the box, and that you repeat this entire procedure 7 times. What is the probability of drawing a green ball exactly 3 times?

8. Suppose a box contains 12 red balls and 8 green balls. Suppose that you pull out a ball at random, check the color, and **do not** replace the ball in the box. You repeat this entire procedure 7 times. What is the probability of drawing a green ball exactly 3 times?

9. A computer chip manufacturer produces a batch of 30 chips. They select three at random (without replacement) to test for defects.

    a) If there is one bad chip in the batch, what is the probability that they find it?

    b) If there are 3 bad chips in the batch, what is the probability that they find one (or more) of them?

    c) If there are 3 bad chips in the batch, what is the probability that they find exactly 2 of them?

10. Suppose a town of 1,000 people has 350 Democrats, 400 Republicans, and 250 independents. A polling company takes a simple random sample of 100 people (a "simple random sample" means it is done without replacement). What is the probability that the sample contains exactly 35 Democrats, 40 Republicans, and 25 independents? Write out your answer in choose notation.

11. You roll a die repeatedly.

a. What is the chance that the first two rolls show the same number of spots?

b. What is the chance of getting exactly 2 sixes in the first 7 rolls?

c. Given that there is exactly 1 six in the first 4 rolls, what is the chance of getting exactly 2 sixes in the first 7 rolls?

d. Given that there is at least 1 six in the first 7 rolls, what is the chance getting exactly 3 sixes in the first 7 rolls?

12. We are investigating a possible relationship between education and partisanship and we collect the following data:

Party/Education	High School or Less	At Least Some College
Democrat	98	202
Republican	100	180

a. If education and partisanship are unrelated, then what is the chance that there would be exactly 100 Republicans with High School or less education out of the 280 Republicans total? Write your answer using choose notation.
b. If education and partisanship are unrelated, then what is the chance that 100 or more Republicans would have High School or less education? Write your answer using choose notation within a sum.
c. Use R to give answers to the first two parts of the question.
d. Is this data particularly unusual if education and partisanship are unrelated?